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Python Pandas

1. What is the shape of the data frame df defined in the below-shown code?

```
import pandas as pd
```

```
data = [{'a': 1, 'b': 2}, {'a': 5, 'b': 10, 'c': 20}]
```

2. Which of the following methods is used to remove duplicates?

Answer : drop_duplicates

3. df = pd.DataFrame(data)

Answer : (2,3)

4. Which of the following attributes or arguments are used to set column names of a data frame?

Answer : columns

5. What is the shape of the data frame df defined in the below-shown code?

```
import pandas as pd
```

```
data = [{'a': 1, 'b': 2}, {'a': 5, 'b': 10, 'c': 20}]
```

```
df = pd.DataFrame(data, columns=['a', 'b'])
```

Answer : (2,2)

6. Which of the following is not a Data Structure of Pandas?

Answer : Dictionary

7. Which of the following cannot be used to create a Data frame?

Answer : A tuple of tuples

8. What is the data type of series s defined in below code?

Answer : object

9. Which of the following argument is used to label the elements of a series?

Answer : index

10. Which of the following expression returns the first two rows of df, defined below?

Answer : Both df[:2] and df.iloc[:2]

11. Which of the following expression is used to add a new column 'C' to a data frame df, with three rows?

Answer : df['C'] = [12, 98, 45]

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12. Which of the following expression is used to delete the column, A from a data frame named df?

Answer : `del df['A']`

13. Which of the following expression returns the second row of df, defined below?

`import pandas`

`df = pd.DataFrame({'A':[34, 78, 54], 'B':[12, 67, 43]}, index=['r1', 'r2', 'r3'])`

Answer : `df.iloc[1]`

14. _____ is used as an argument of the readcsv method to make data of a specific column as an index.

Answer : `index_col`

15. Which of the following method is used to read data from excel files?

Answer :. `read_excel`

16. Which of the following is used as argument of read_csv method to treat data of specific columns as dates?

Answer : `parse_dates`

17. State whether the following statement is true or false? The read_csv method, by default, reads all blank lines of an input CSV file.

Answer : False

18. Which of the following method is used to write a data frame data to an output CSV file?

Answer : `to_csv`

19. State whether the following statement is true or false? The read_csv method can read multiple columns of an input file as indexes.

Answer : True

20. Which of the following is used as an argument of read_csv method to skip first n lines of an input CSV file?

Answer : `skiprows`

21. What is the length of DatetimeIndex object created with the below expression?

`pd.date_range('11-Sep-2017', '17-Sep-2017', freq='2D')`

Answer : 4

22. Which of the following expressions are used to check if each element of a series s is present in the list of elements [67, 32]. Series s is defined as shown below.

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```
s = pd.Series([99, 32, 67],list('abc'))
```

Answer : `s.isin([67, 32])`

23. Which of the following method is used to convert a list of dates like strings into datetime objects?

Answer : `to_datetime`

24. What is the output of the following code?

```
import pandas as pd

d = pd.date_range('11-Sep-2017', '17-Sep-2017', freq='2D')

len(d[d.isin(pd.to_datetime(['12-09-2017', '15-09-2017']))])
```

Answer : 1

25. What is the length of Period Index object created from the expression `pd.period_range('11-Sep-2017', '17-Sep-2017', freq='M')`?

Answer : 1

26. Which of the following method is used to fill null values with a default value?

Answer : `fillna`

27. Which of the following argument values are allowed for the method argument of `fillna`?

Answer : All

28. By default, missing values in any data set are read as

Answer : NaN

29. Which of the following methods is used to remove duplicates?

Answer : `drop_duplicates`

30. Unrecognized datetime value is treated as _____.

Answer : NaT

31. Which of the following method of pandas is used to check if each value is a null or not?

Answer : `isnull`

32. Which of the following method is used to eliminate rows with null values?

Answer : `dropna`

33. Consider a data frame `df` with columns ['A', 'B', 'C', 'D'] and rows ['r1', 'r2', 'r3']. What does the expression `df[lambda x : x.index.str.endswith('3')]` do?

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Answer : Filters the row labelled r3

34. Consider a data frame df with columns ['A', 'B', 'C', 'D'] and rows ['r1', 'r2', 'r3'], Which of the following expression is used to extract columns 'C' and 'D'?

Answer : df.loc[:, lambda x : x.columns.isin(['C', 'D'])]

35. Consider a data frame df with 10 rows and index ['r1', 'r2', 'r3', 'row4', 'row5', 'row6', 'r7', 'r8', 'r9', 'row10']. What does the expression g = df.groupby(df.index.str.len()) do?

Answer : Groups df based on length of each index value

36. Consider a data frame df with columns ['A', 'B', 'C', 'D'] and rows ['r1', 'r2', 'r3']. Which of the following expression filters the rows whose column B values are greater than 45 and column 'C' values are less than 30?

Answer : df.loc[(df.B > 45) & (df.C < 30)]

37. Which of the following methods is used to group data of a data frame, based on specific columns?

Answer : groupby

38. Which of the following method can be applied on a groupby object to get the group details?

Answer : groups

39. What does the expression df.iloc[:, lambda x : [0,3]] do? Consider a data frame df with columns ['A', 'B', 'C', 'D'] and rows ['r1', 'r2', 'r3'].

Answer : Selects Columns 'A', and 'D'

40. Consider a data frame df with 10 rows and index ['r1', 'r2', 'r3', 'row4', 'row5', 'row6', 'r7', 'r8', 'r9', 'row10']. What does the aggregate method shown in below code do?

```
g = df.groupby(df.index.str.len())
```

```
g.aggregate({'A':len, 'B':np.sum})
```

Answer : Computes length of column A and Sum of Column B values of each group

41. Consider a data frame df with 10 rows and index ['r1', 'r2', 'r3', 'row4', 'row5', 'row6', 'r7', 'r8', 'r9', 'row10']. How many rows are obtained after executing the below expressions

```
g = df.groupby(df.index.str.len())
```

```
g.filter(lambda x: len(x) > 1)
```

Answer : 9

42. Which of the following method is used to concatenate two or more data frames?

Answer : concat

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43. Which of the following argument is used to set the key to be used for merging two data frames?

Answer : on

44. Which of the following are allowed values of the argument how of merge method?

Answer : All of the options

45. Which argument is used to override the existing column names, while using concat method?

Answer : keys

46. Which of the following argument is used to ignore the index while concatenating two data frames?

Answer : ignore_index

47. What is the shape of d defined in below code?

```
import pandas as pd  
s1 = pd.Series([0, 1, 2, 3])  
s2 = pd.Series([0, 1, 2, 3])  
s3 = pd.Series([0, 1, 4, 5])  
d = pd.concat([s1, s2, s3], axis=1)
```

Answer : (4,3)

48. Which of the following expressions are used to check if each element of a series s is present in the list of elements [67, 32]. Series s is defined as shown below.

```
s = pd.Series([99, 32, 67], list('abc'))
```

Answer : s.isin([67,32])

49. What is the output of the expression 'b' in s, where s is the series defined as shown below?

```
s = pd.Series([89.2, 76.4, 98.2, 75.9], index=list('abcd'))
```

Answer : True

50. What is the shape of the data frame df defined in the below shown code?

```
import pandas as pd  
data = [{'a': 1, 'b': 2}, {'a': 5, 'b': 10, 'c': 20}]  
df = pd.DataFrame(data, columns=['a', 'b'])
```

Answer : (2,2)

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51. Which of the following expression returns data of column B of data frame df, defined below?
Which of the following expression returns the data of column B in data frame df, defined below.
import pandas as pd

```
df = pd.DataFrame({'A':[34, 78, 54], 'B':[12, 67, 43]}, index=['r1', 'r2', 'r3'])
```

Answer : df.B

52. Which of the following expression returns the first two rows of df, defined below?

import pandas as pd

```
df = pd.DataFrame({'A':[34, 78, 54], 'B':[12, 67, 43]}, index=['r1', 'r2', 'r3'])
```

Answer : Both df[:2] and df.iloc[:2]

53. Which of the following expression returns last two rows of df, defined below?

import pandas as pd

```
df = pd.DataFrame({'A':[34, 78, 54], 'B':[12, 67, 43]}, index=['r1', 'r2', 'r3'])
```

Answer : df.loc['r2':'r3']

54. What does the expression df.loc['r4'] = [67, 78] do for the data frame df, defined below?

```
df = pd.DataFrame({'A':[34, 78, 54], 'B':[12, 67, 43]}, index=['r1', 'r2', 'r3'])
```

Answer : Add a new row

55. Which of the following method is used to read data from excel files?

Answer : read_excel

56. What does the expression d + pd.Timedelta('1 days 2 hours') do to DatetimeIndex object d, defined below? d = pd.date_range('11-Sep-2017', '17-Sep-2017', freq='2D')

Answer : Increases each datetime value by 1 day and 2 hours

57. Which of the following method is used to convert a list of dates like strings into datetime objects?

Answer : to_datetime

58. By default, missing values in any data set are read as

Answer : NaN

59. Which of the following argument values are allowed for the method argument of fillna?

Answer : All the options

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60. Consider a data frame df with columns ['A', 'B', 'C', 'D'] and rows ['r1', 'r2', 'r3']. Which of the following expression filters those rows whose column B values are greater than 45?

Answer : `df[df.B > 45]`

61. Consider a data frame df with 10 rows and index ['r1', 'r2', 'r3', 'row4', 'row5', 'row6', 'r7', 'r8', 'r9', 'row10']. What does the expression `g = df.groupby(df.index.str.len())` do?

Answer : Data frames cannot be grouped by index values. Hence it results in Error.

62. What is the output of the following code?

```
import pandas as pd

s = pd.Series([89.2, 76.4, 98.2, 75.9], index=list('abcd'))

print(s[['c', 'a']])
```

Answer : `c 98.2 a 89.2 dtype: float64`